Logic (week 4)

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To DNF (ex. 4.42)

$$B \wedge ((A \wedge B) \wedge ((A \vee B) \vee (B \wedge C)))$$

To DNF (ex. 4.42)

$$B \wedge ((A \wedge B) \wedge ((A \vee B) \vee (B \wedge C))) \quad \Leftrightarrow \quad (A\wedge)$$
$$(B \wedge (A \wedge B)) \wedge ((A \vee B) \vee (B \wedge C))$$

To DNF (ex. 4.42)

$$
\begin{aligned}
B \wedge ((A \wedge B) \wedge ((A \vee B) \vee (B \wedge C))) &\Leftrightarrow (A\wedge) \\
(B \wedge (A \wedge B)) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (C\wedge) \\
(B \wedge (B \wedge A)) \wedge ((A \vee B) \vee (B \wedge C)) &
\end{aligned}
$$

To DNF (ex. 4.42)

$$
\begin{aligned}
B \wedge ((A \wedge B) \wedge ((A \vee B) \vee (B \wedge C))) &\Leftrightarrow (A\wedge) \\
(B \wedge (A \wedge B)) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (C\wedge) \\
(B \wedge (B \wedge A)) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (A\wedge) \\
((B \wedge B) \wedge A) \wedge ((A \vee B) \vee (B \wedge C))
\end{aligned}
$$

To DNF (ex. 4.42)

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\begin{aligned}
B \wedge ((A \wedge B) \wedge ((A \vee B) \vee (B \wedge C))) &\Leftrightarrow (A\wedge)\\
(B \wedge (A \wedge B)) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (C\wedge)\\
(B \wedge (B \wedge A)) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (A\wedge)\\
((B \wedge B) \wedge A) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (I\wedge)\\
(B \wedge A) \wedge ((A \vee B) \vee (B \wedge C))
\end{aligned}
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(B \wedge (B \wedge A)) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (A\wedge)\\
((B \wedge B) \wedge A) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (I\wedge)\\
(B \wedge A) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (\text{left-}D\wedge)\\
((B \wedge A) \wedge (A \vee B)) \vee ((B \wedge A) \wedge (B \wedge C))
\end{aligned}
$$

To DNF (ex. 4.42)

$$
\begin{aligned}
B \wedge ((A \wedge B) \wedge ((A \vee B) \vee (B \wedge C))) &\Leftrightarrow (A\wedge)\\
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((B \wedge B) \wedge A) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (I\wedge)\\
(B \wedge A) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (\text{left-}D\wedge)\\
((B \wedge A) \wedge (A \vee B)) \vee ((B \wedge A) \wedge (B \wedge C)) &\Leftrightarrow (\text{left-}D\wedge)\\
(((B \wedge A) \wedge A) \vee ((B \wedge A) \wedge B)) \vee ((B \wedge A) \wedge (B \wedge C)) &
\end{aligned}
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To DNF (ex. 4.42)

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(B \wedge (B \wedge A)) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (A\wedge)\\
((B \wedge B) \wedge A) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (I\wedge)\\
(B \wedge A) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (\text{left-}D\wedge)\\
((B \wedge A) \wedge (A \vee B)) \vee ((B \wedge A) \wedge (B \wedge C)) &\Leftrightarrow (\text{left-}D\wedge)\\
(((B \wedge A) \wedge A) \vee ((B \wedge A) \wedge B)) \vee ((B \wedge A) \wedge (B \wedge C))
\end{aligned}
$$

This is already in DNF. With the ACI-laws for conjunction, it can be simplified:

$$(B \wedge A) \vee (A \wedge B \wedge C)$$

(still in DNF)

To DNF (ex. 4.42)

$$
\begin{aligned}
B \wedge ((A \wedge B) \wedge ((A \vee B) \vee (B \wedge C))) &\Leftrightarrow (A\wedge)\\
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(B \wedge (B \wedge A)) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (A\wedge)\\
((B \wedge B) \wedge A) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (I\wedge)\\
(B \wedge A) \wedge ((A \vee B) \vee (B \wedge C)) &\Leftrightarrow (\text{left-}D\wedge)\\
((B \wedge A) \wedge (A \vee B)) \vee ((B \wedge A) \wedge (B \wedge C)) &\Leftrightarrow (\text{left-}D\wedge)\\
(((B \wedge A) \wedge A) \vee ((B \wedge A) \wedge B)) \vee ((B \wedge A) \wedge (B \wedge C))
\end{aligned}
$$

This is already in DNF. With the ACI-laws for conjunction, it can be simplified:

$$(B \wedge A) \vee (A \wedge B \wedge C)$$

(still in DNF)

In fact, this is equivalent to $A \wedge B$ (which would still be in DNF), but you cannot get there with only the rules we've discussed.

To CNF (syllabus 9.2c)

$$\neg((P \to Q) \wedge (R \to S))$$

To CNF (syllabus 9.2c)

$$\neg((P \to Q) \wedge (R \to S)) \quad \Leftrightarrow \quad (\text{Def} \to)$$
$$\neg((\neg P \vee Q) \wedge (\neg R \vee S))$$

To CNF (syllabus 9.2c)

$$
\begin{aligned}
\neg((P \to Q) \wedge (R \to S)) \quad &\Leftrightarrow \quad (\text{Def} \to) \\
\neg((\neg P \vee Q) \wedge (\neg R \vee S)) \quad &\Leftrightarrow \quad (\text{De Morgan}) \\
\neg(\neg P \vee Q) \vee \neg(\neg R \vee S) \quad &
\end{aligned}
$$

To CNF (syllabus 9.2c)

$$
\begin{aligned}
\neg((P \to Q) \wedge (R \to S)) &\quad \Leftrightarrow \quad (\text{Def} \to) \\
\neg((\neg P \vee Q) \wedge (\neg R \vee S)) &\quad \Leftrightarrow \quad (\text{De Morgan}) \\
\neg(\neg P \vee Q) \vee \neg(\neg R \vee S) &\quad \Leftrightarrow \quad (\text{De Morgan}) \\
(\neg\neg P \wedge \neg Q) \vee (\neg\neg R \wedge \neg S)
\end{aligned}
$$

To CNF (syllabus 9.2c)

$$
\begin{aligned}
\neg((P \to Q) \wedge (R \to S)) &\quad \Leftrightarrow \quad (\text{Def} \to) \\
\neg((\neg P \vee Q) \wedge (\neg R \vee S)) &\quad \Leftrightarrow \quad (\text{De Morgan}) \\
\neg(\neg P \vee Q) \vee \neg(\neg R \vee S) &\quad \Leftrightarrow \quad (\text{De Morgan}) \\
(\neg\neg P \wedge \neg Q) \vee (\neg\neg R \wedge \neg S) &\quad \Leftrightarrow \quad (\text{double-neg}) \\
(P \wedge \neg Q) \vee (R \wedge \neg S)
\end{aligned}
$$

To CNF (syllabus 9.2c)

$$
\begin{aligned}
\neg((P \to Q) \wedge (R \to S)) &\quad \Leftrightarrow \quad (\text{Def} \to) \\
\neg((\neg P \vee Q) \wedge (\neg R \vee S)) &\quad \Leftrightarrow \quad (\text{De Morgan}) \\
\neg(\neg P \vee Q) \vee \neg(\neg R \vee S) &\quad \Leftrightarrow \quad (\text{De Morgan}) \\
(\neg\neg P \wedge \neg Q) \vee (\neg\neg R \wedge \neg S) &\quad \Leftrightarrow \quad (\text{double-neg}) \\
(P \wedge \neg Q) \vee (R \wedge \neg S) &\quad \Leftrightarrow \quad (\text{left-}D\vee) \\
((P \wedge \neg Q) \vee R) \wedge ((P \wedge \neg Q) \vee \neg S)
\end{aligned}
$$

To CNF (syllabus 9.2c)

$$
\begin{aligned}
\neg((P \to Q) \wedge (R \to S)) &\Leftrightarrow (\text{Def} \to) \\
\neg((\neg P \vee Q) \wedge (\neg R \vee S)) &\Leftrightarrow (\text{De Morgan}) \\
\neg(\neg P \vee Q) \vee \neg(\neg R \vee S) &\Leftrightarrow (\text{De Morgan}) \\
(\neg\neg P \wedge \neg Q) \vee (\neg\neg R \wedge \neg S) &\Leftrightarrow (\text{double-neg}) \\
(P \wedge \neg Q) \vee (R \wedge \neg S) &\Leftrightarrow (\text{left-}D\vee) \\
((P \wedge \neg Q) \vee R) \wedge ((P \wedge \neg Q) \vee \neg S) &\Leftrightarrow (\text{right-}D\vee) \\
((P \vee R) \wedge (\neg Q \vee R)) \wedge ((P \vee \neg S) \wedge (\neg Q \vee \neg S))
\end{aligned}
$$

To CNF (syllabus 9.2c)

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\begin{aligned}
\neg((P \to Q) \wedge (R \to S)) &\Leftrightarrow (\text{Def} \to) \\
\neg((\neg P \vee Q) \wedge (\neg R \vee S)) &\Leftrightarrow (\text{De Morgan}) \\
\neg(\neg P \vee Q) \vee \neg(\neg R \vee S) &\Leftrightarrow (\text{De Morgan}) \\
(\neg\neg P \wedge \neg Q) \vee (\neg\neg R \wedge \neg S) &\Leftrightarrow (\text{double-neg}) \\
(P \wedge \neg Q) \vee (R \wedge \neg S) &\Leftrightarrow (\text{left-}D\vee) \\
((P \wedge \neg Q) \vee R) \wedge ((P \wedge \neg Q) \vee \neg S) &\Leftrightarrow (\text{right-}D\vee) \\
((P \vee R) \wedge (\neg Q \vee R)) \wedge ((P \vee \neg S) \wedge (\neg Q \vee \neg S))
\end{aligned}
$$

This is in CNF.